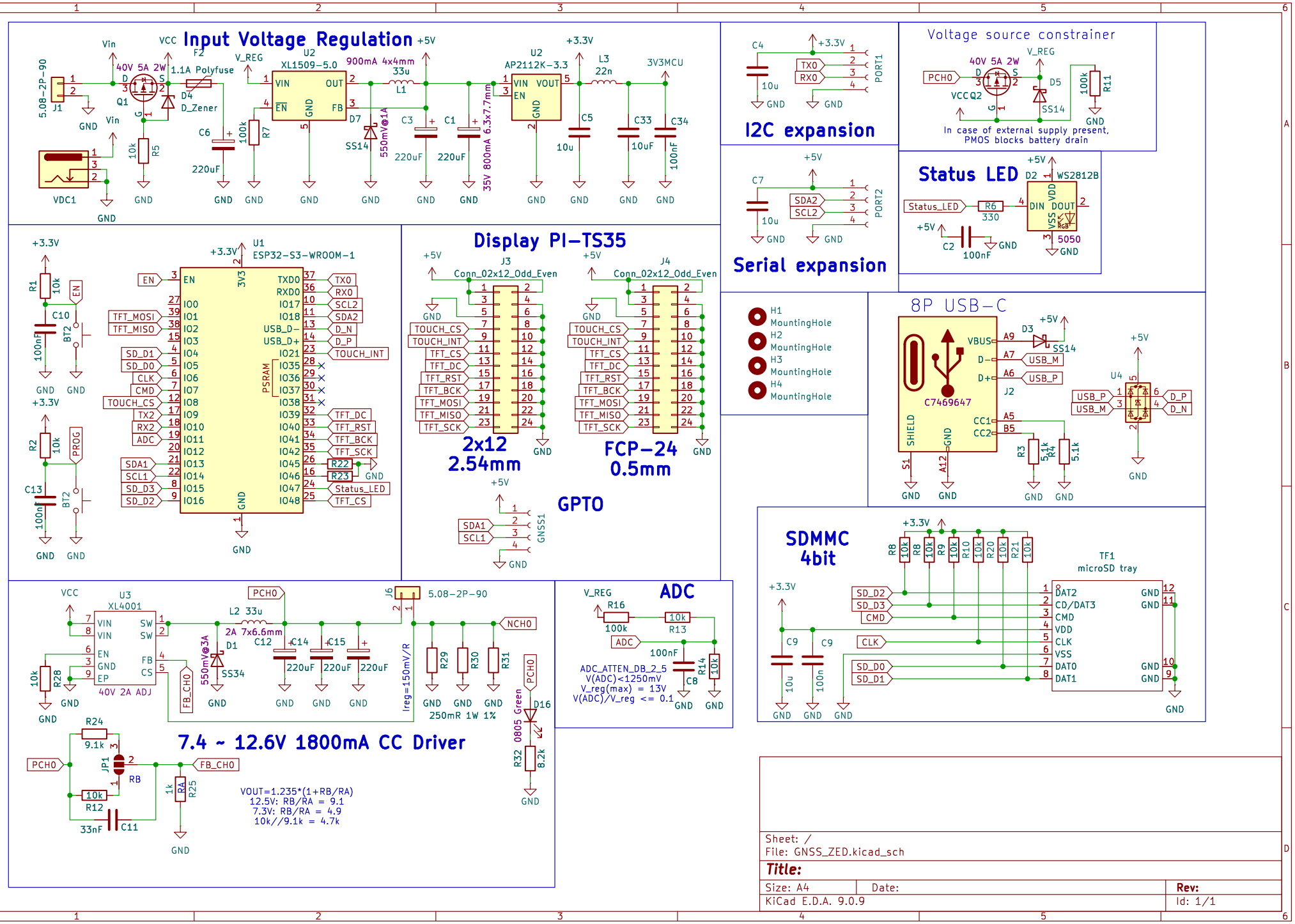
Input Voltage Regulation
Vin
VCC
F2
1.1A Polyfuse
40V 5A 2W
Q1
D4
D_Zener
J1
5.08-2P-90
VDC1
R5
10k
C6
220uF
V_REG
U2
XL1509-5.0
VIN
OUT
EN
GND
FB
R7
100k
D7
SS14
550mV@1A
900mA 4x4mm
L1
33u
C3
220uF
C1
220uF
+5V
U2
AP2112K-3.3
VIN
VOUT
EN
GND
35V 800mA 6.3x7.7mm
+3.3V
C5
10u
L3
22n
C33
10uF
C34
100nF
3V3MCU
U1
ESP32-S3-WROOM-1
EN
3V3
TXD0
RXD0
IO17
IO18
USB_D-
USB_D+
IO21
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
PSRAM
TFT_MOSI
TFT_MISO
SD_D1
SD_D0
CLK
CMD
TOUCH_CS
TX2
RX2
ADC
SDA1
SCL1
SD_D3
SD_D2
TX0
RX0
SCL2
SDA2
D_N
D_P
TOUCH_INT
TFT_DC
TFT_RST
TFT_BCK
TFT_SCK
R22
R23
Status_LED
TFT_CS
R1
10k
C10
100nF
BT2
R2
10k
PROG
C13
100nF
Display PI-TS35
J3
Conn_02x12_Odd_Even
J4
Conn_02x12_Odd_Even
TOUCH_CS
TOUCH_INT
TFT_CS
TFT_DC
TFT_RST
TFT_BCK
TFT_MOSI
TFT_MISO
TFT_SCK
2x12
2.54mm
FCP-24
0.5mm
GPTO
SDA1
SCL1
GNSS1
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
I2C expansion
C4
10u
TX0
RX0
PORT1
Serial expansion
C7
10u
SDA2
SCL2
PORT2
Voltage source constrainer
PCH0
40V 5A 2W
Q2
V_REG
D5
SS14
100k
R11
In case of external supply present,
PMOS blocks battery drain
Status LED
D2
WS2812B
Status_LED
R6
330
DIN
DOUT
VDD
VSS
5050
C2
100nF
8P USB-C
C7469647
VBUS
D-
D+
SHIELD
GND
A9
A7
A6
A5
B5
A12
S1
D3
SS14
USB_M
USB_P
CC1
CC2
J2
R3
5.1k
R4
5.1k
U4
D_P
D_N
SDMMC
4bit
TF1
microSD tray
SD_D2
SD_D3
CMD
CLK
SD_D0
SD_D1
DAT2
CD/DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
R8
R9
R10
R20
R21
C9
10u
C9
100n
U3
XL4001
VIN
SW
EN
GND
FB
CS
EP
40V 2A ADJ
R28
10k
FB_CH0
PCH0
L2 33u
2A 7x6.6mm
D1
SS34
550mV@3A
C12
C14
C15
220uF
220uF
220uF
J6
5.08-2P-90
NCH0
Ireg=150mV/R
R29
R30
R31
250mR 1W 1%
D16
0805 Green
R32
8.2k
7.4 ~ 12.6V 1800mA CC Driver
R24
9.1k
JP1
RB
R12
10k
33nF
C11
1k
RA
R25
VOUT=1.235*(1+RB/RA)
12.5V: RB/RA = 9.1
7.3V: RB/RA = 4.9
10k//9.1k = 4.7k
ADC
V_REG
R16
100k
R13
10k
ADC
100nF
C8
R14
10k
ADC_ATTEN_DB_2_5
V(ADC)<1250mV
V_reg(max) = 13V
V(ADC)/V_reg <= 0.1
Sheet: /
File: GNSS_ZED.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.9
Id: 1/1